Vzorce pro diferenční a sumační počet

$$\Delta f_t := f_{t+1} - f_t, \qquad \sum f_t = F_t + C \quad \Longleftrightarrow \quad \Delta F_t = f_t.$$

Diference

$$\begin{aligned}
\Delta c &= 0,\\
\Delta t &= 1,\\
\Delta t^2 &= 2t+1,\\
\Delta t^3 &= 3t^2+3t+1,\\
\Delta(t^m) &= \sum_{j=1}^{m} \binom{m}{j} t^{m-j},\\
\Delta a^t &= (a-1)a^t,\\
\Delta e^{\alpha t} &= (e^{\alpha}-1)e^{\alpha t}.
\end{aligned}$$

$$\boxed{\Delta t^{(r)} = r\, t^{(r-1)}}$$

kde $t^{(r)} = t(t-1)\cdots(t-r+1)$, $\qquad t^{(0)} = 1$.

$$\begin{aligned}
\Delta\tbinom{t}{r} &= \tbinom{t}{r-1},\\
\Delta \sin(\alpha t) &= 2\sin\frac{\alpha}{2}\,\cos\left(\alpha t+\frac{\alpha}{2}\right),\\
\Delta \cos(\alpha t) &= -2\sin\frac{\alpha}{2}\,\sin\left(\alpha t+\frac{\alpha}{2}\right),\\
\Delta \sin(\alpha t) &= (\cos\alpha - 1)\sin(\alpha t) + \sin\alpha\cos(\alpha t),\\
\Delta \cos(\alpha t) &= (\cos\alpha - 1)\cos(\alpha t) - \sin\alpha\sin(\alpha t),\\
\Delta \operatorname{tg}(\alpha t) &= \frac{\sin\alpha}{\cos(\alpha t)\cos(\alpha(t+1))},\\
\Delta \ln t &= \ln\left(1+\frac{1}{t}\right),\\
\Delta \log_a t &= \log_a\left(1+\frac{1}{t}\right),\\
\Delta \ln\Gamma(t) &= \ln t,\\
\Delta H_t &= \frac{1}{t+1}.
\end{aligned}$$

Pro $H_t = \sum_{j=1}^{t} 1/j$, $H_0 = 0$.

$$\begin{aligned}
\Delta \operatorname{arctg} t &= \operatorname{arctg}\frac{1}{t^2+t+1},\\
\Delta \arcsin t &= \arcsin(t+1) - \arcsin t.
\end{aligned}$$

Neurčité sumy

$$\begin{aligned}
\sum 1 &= t + C,\\
\sum t &= \frac{t^{(2)}}{2} + C = \frac{t(t-1)}{2} + C,\\
\sum t^{(r)} &= \frac{t^{(r+1)}}{r+1} + C, \qquad r \neq -1,\\
\sum \tbinom{t}{r} &= \tbinom{t}{r+1} + C,\\
\sum a^t &= \frac{a^t}{a-1} + C, \qquad a \neq 1,\\
\sum e^{\alpha t} &= \frac{e^{\alpha t}}{e^{\alpha}-1} + C, \qquad \alpha \neq 0.
\end{aligned}$$

$$\begin{aligned}
\sum \sin(\alpha t) &= -\frac{\cos\left(\alpha t - \frac{\alpha}{2}\right)}{2\sin\frac{\alpha}{2}} + C,\\
\sum \cos(\alpha t) &= \frac{\sin\left(\alpha t - \frac{\alpha}{2}\right)}{2\sin\frac{\alpha}{2}} + C, \qquad \text{pro } \sin(\alpha/2) \neq 0.
\end{aligned}$$

$$\begin{aligned}
\sum \ln t &= \ln\Gamma(t) + C,\\
\sum \ln\left(1+\frac{1}{t}\right) &= \ln t + C,\\
\sum \frac{1}{t+1} &= H_t + C.
\end{aligned}$$

Některé mocninné součty:

$$\begin{aligned}
\sum t^2 &= \frac{t(t-1)(2t-1)}{6} + C,\\
\sum t^3 &= \frac{t^2(t-1)^2}{4} + C,\\
\sum t^4 &= \frac{t(t-1)(2t-1)(3t^2-3t-1)}{30} + C.
\end{aligned}$$

Stirlingova čísla prvního druhu.

$$\begin{bmatrix} m \\ r \end{bmatrix} = \begin{bmatrix} m-1 \\ r-1 \end{bmatrix} + (m-1)\begin{bmatrix} m-1 \\ r \end{bmatrix},$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1, \qquad \begin{bmatrix} m \\ 0 \end{bmatrix} = 0 \ (m>0), \qquad \begin{bmatrix} 0 \\ r \end{bmatrix} = 0 \ (r>0).$$

$$t^{(m)} = \sum_{r=0}^{m} (-1)^{m-r} \begin{bmatrix} m \\ r \end{bmatrix} t^r.$$

Stirlingova čísla druhého druhu.

$$\left\{ {m \atop r} \right\} = \left\{ {m-1 \atop r-1} \right\} + r\left\{ {m-1 \atop r} \right\},$$

$$\left\{ {0 \atop 0} \right\} = 1, \qquad \left\{ {m \atop 0} \right\} = 0 \ (m>0), \qquad \left\{ {0 \atop r} \right\} = 0 \ (r>0).$$

$$t^m = \sum_{r=0}^{m} \left\{ {m \atop r} \right\} t^{(r)}, \qquad \sum t^m = \sum_{r=0}^{m} \left\{ {m \atop r} \right\} \frac{t^{(r+1)}}{r+1} + C.$$

Pravidla pro diference

$$\begin{aligned}
\Delta(u_t \pm v_t) &= \Delta u_t \pm \Delta v_t,\\
\Delta(cu_t) &= c\,\Delta u_t,\\
\Delta(u_t v_t) &= (\Delta u_t)v_t + u_{t+1}\Delta v_t,\\
&= u_t\Delta v_t + v_{t+1}\Delta u_t,\\
\Delta\left(\frac{u_t}{v_t}\right) &= \frac{v_t\Delta u_t - u_t\Delta v_t}{v_t v_{t+1}}.
\end{aligned}$$

Vyšší diference:

$$\Delta^n u_t = \sum_{j=0}^{n} (-1)^{n-j}\binom{n}{j} u_{t+j}$$

Určitý součet a sumace per partes

Je-li $\Delta F_t = f_t$, pak

$$\boxed{\sum_{t=m}^{n-1} f_t = F_n - F_m}$$

a zejména

$$\sum_{t=m}^{n-1} \Delta u_t = u_n - u_m.$$

$$\sum_{t=m}^{n-1} u_t\,\Delta v_t = u_n v_n - u_m v_m - \sum_{t=m}^{n-1} v_{t+1}\,\Delta u_t.$$